

Assignment 1: Part C (Modelling)

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Problem Description

Autonomous rovers sent to Mars must be able to decide whether the patches of terrain surrounding it are traversable or non-traversable, in real time. There are many computer vision systems that have been developed to help autonomous rovers in this aspect. However, one of the biggest problems is still the lack of training data (clear images of Martian terrain as it would appear from a rover's camera). In order to explore this field further and confirm whether transfer learning through mixed-domain training could benefit vision systems, we proposed the following refined research question:

How accurately can a CNN, trained on the AI4Mars dataset, classify 320×320 rover-view patches, and how does co-training with the terrestrial Freiburg-Forest dataset affect that accuracy?

Input Data

Source	Format	Size (train / test)	Bands	Label mask values ~> class
AI4Mars v0.1 (JPL 2020)	1024 × 1024 RGB JPG	16 064 / 322	visible	0, 1 = <i>safe</i> ; 2, 3 = <i>hazard</i> ; 255 = <i>ignore</i>
Freiburg Forest (Valada <i>et al.</i> 2016)	Varied Dimension RGB JPG	230 / 136	visible	(170,170,170) = <i>road</i> (safe); vegetation/obstacle colours = <i>hazard</i>

Output

- Binary label at patch level:
 - 0 = safe
 - 1 = hazard

Summary of Key Features

- Class imbalance
 - Ratio of mars images is greater than earth images
 - Ratio of safe images is greater than hazardous images
- Spectral shift
 - Mars images are in a grey hue distribution
 - Earth images are in a colour-coded distribution
- Texture variance

- Grey-level standard deviation
- Edge density

Data Pre-Processing

Step	Rationale
Mask images are converted to binary using <i>vectorized</i> np.isin() mapping for AI4Mars and colour matching for Freiburg.	Promotes fast 8-bit operation without needing loops.
Images are centre-cropped to 320 px (Mars-specific camera FOV ≈ 320).	Creates a uniform input and removes rover hardware & sky artefacts.
Images are normalized to RGB / 255.	Is a best practice for Pytorch and avoids additional scaling.
Albumentations is used to apply HorizontalFlip, Rotate ($\pm 15^\circ$), mild ColorJitter.	Increases hazard diversity but keeps rotations limited to preserve horizon orientation.
Class-balanced weighting is calculated pos_weight = num_safe / num_hazard to be used in BCE loss.	Compensates for hazard scarcity instead of oversampling.
AMP (mixed precision) and pin_memory is used for GPU throughput.	Reduces the VRAM usage on GPU.

Since there were no missing values, no imputation strategies were necessary.

Model Selection

Model	Motivation	Params	Input
Logistic regression (LR) on 5-D handcrafted vector.	Provides an analytical baseline and is easily interpretable.	6	mean RGB + grey std + edge density
Mini-CNN (3 $\times$ Conv $\leadsto$ GAP $\leadsto$ FC).	Is a lightweight, end-to-end learner with < 70 k params and fits 320 $\times$ 320 images.	69 k	raw RGB 320x320
MobileNet-v2 head (encoder frozen, GAP + FC).	Has a modern lightweight backbone and tests the benefit of deeper features without having a huge training cost.	2.2 M trainable	raw RGB 224x224

UNet was initially selected to be a 3rd trial model but was abandoned for creating Out Of Memory errors on the GPU. Instead, MobileNet was retained as a 3rd trial model. However, input images had to be cropped to 224 px instead of 320 to better suit the MobileNet architecture.

Model Refinement

Procedure	LR	Mini-CNN	MobileNet-v2
Optimiser	n/a	Adam with learning rate = 1 e-3	Adam with learning rate = 1 e-3
Batch	n/a	32	8 (224 px)
Epochs / early-stop	n/a	3 + patience 3 (validated with Balanced Accuracy)	3 + patience 3 (validated with Balanced Accuracy)
Loss	Balanced Weight Vector	BCEWithLogits(pos_weight)	BCEWithLogits(pos_weight)
Validation split	Fixed Mars test set (322 patches)	Fixed Mars test set (322 patches)	Fixed Mars test set (322 patches)
Mixed precision	n/a	AMP autocast + grad-scaler	AMP autocast + grad-scaler

Performance Comparisons

Model	Mars-only	Mars + Earth	Δ (pp)
Logistic regression	0.813	0.826	+1.3
Mini-CNN	0.701	0.717	+1.6
MobileNet-v2	0.818	0.864	+4.6

*Metric = balanced accuracy (BA) on Mars test set

**Chosen to account for imbalanced classes

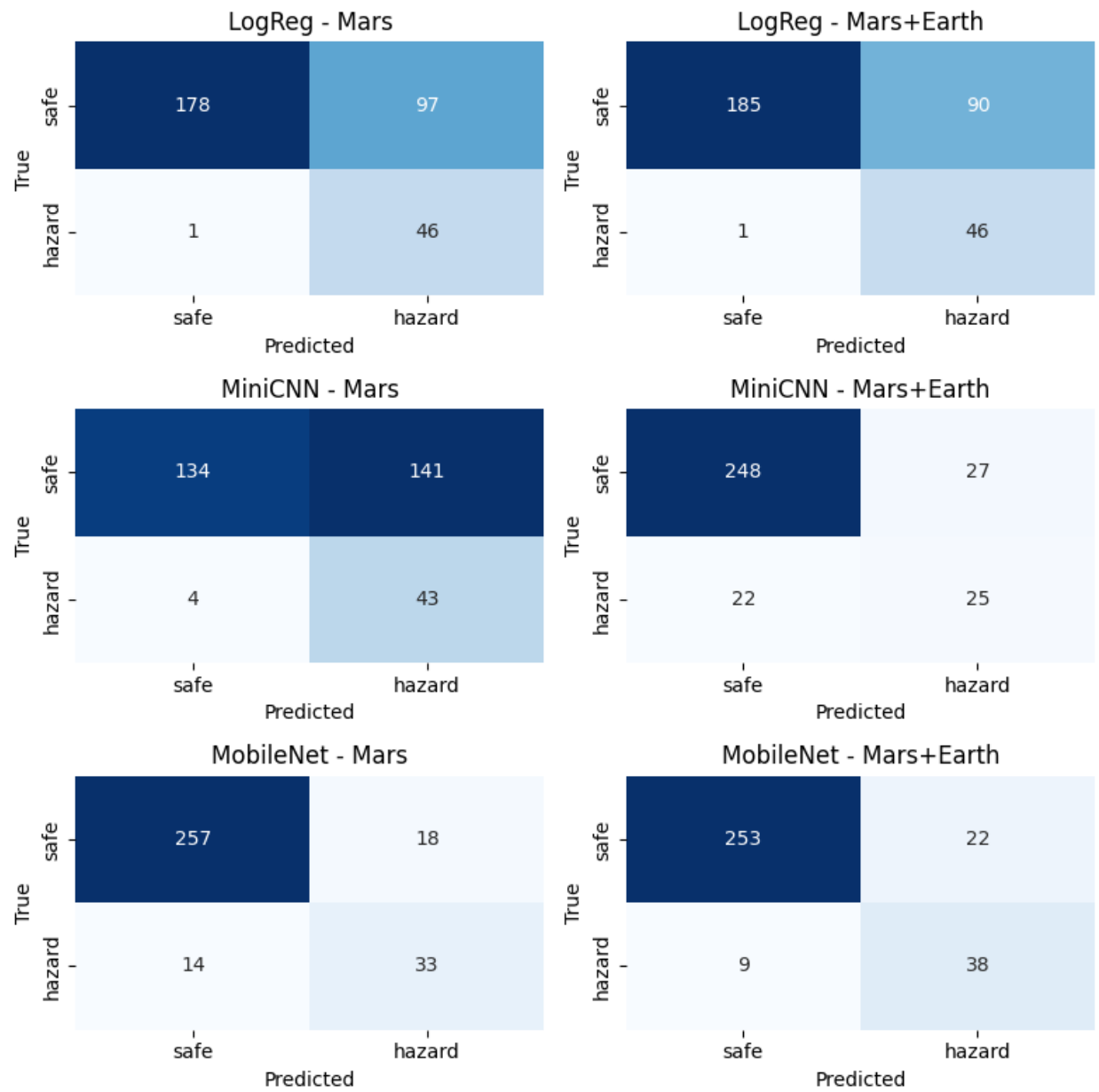


Figure 1 – Confusion Matrices detailing True vs Predicted Labels

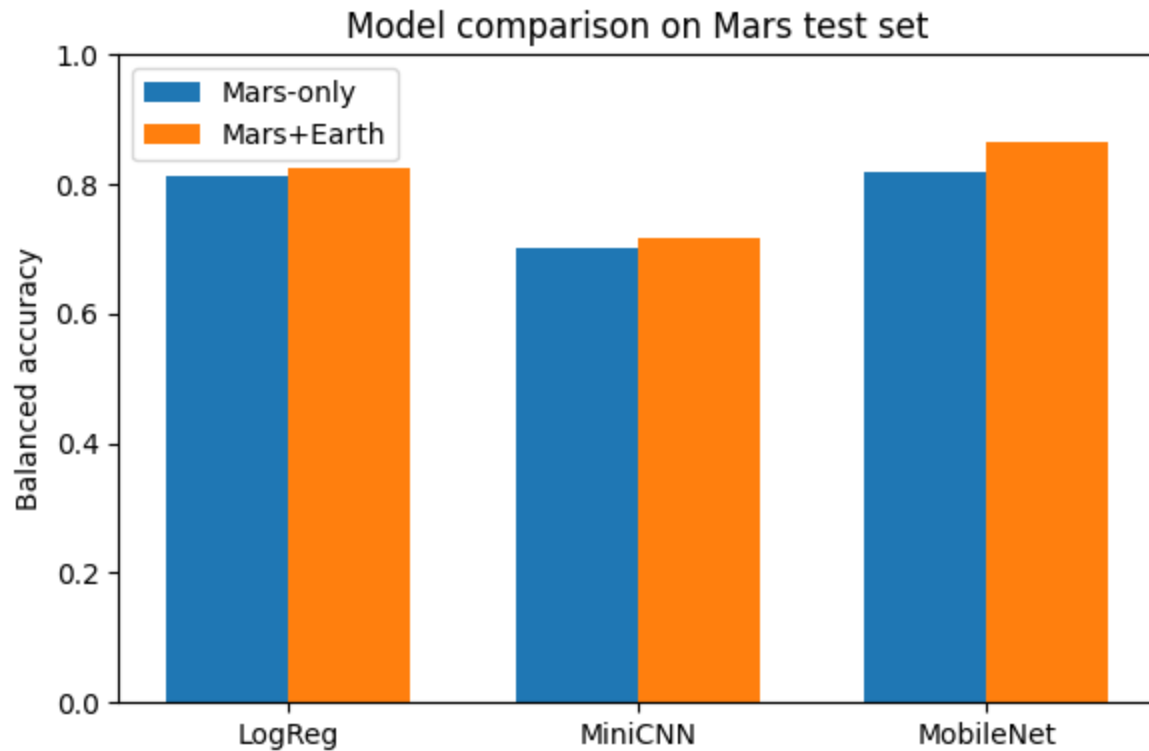


Figure 2 – Bar Chart summarizing the Balanced Accuracies across models and datasets

Results Interpretation

- Logistic Regression
 - Our Learning Rate is already at a respectable 82.6% with the combined dataset
 - The Earth co-training improved the safe-recall slightly (178 ~> 185 TN) without giving us a hazard penalty (TP stays as 46)
- Mini-CNN
 - The Mini-CNN (Mars-only) struggled to learn both colour and texture cues (safe-recall 0.49)
 - Adding the 230 terrestrial paths doubled safe-precision and boosted overall BA by +1.6 pp
 - This is evidence of useful spectral diversity transfer
- MobileNet-v2
 - MobileNet-v2 emerged as the best model having 0.864 BA with co-training, mainly by cutting Mars FN from 14 → 9 (as you can see in Figure 1)
 - We can conclude that deeper pre-trained filters generalize colour shift better, and GAP over 224 px subsamples noise

From these observations we can say the model of choice is MobileNet-v2. It has the highest BA and a consistent hazard recall (0.81), while also maintaining an inference of < 5 ms on GPU (this would be suitable for on-board rover latency).

- Remaining Gaps in Implementation:
 - Hazard representation is still tiny (only 47 patches)
 - Synthetic oversampling or contrastive self-supervision on un-labelled Freiburg RAW images could help
 - Domain shifts (like dust or lighting) could be tackled with style transfer augmentation

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Appendix

Mars-only Logistic Regression					
	precision	recall	f1-score	support	
safe	0.994	0.647	0.784	275	
hazard	0.322	0.979	0.484	47	
accuracy			0.696	322	
macro avg	0.658	0.813	0.634	322	
weighted avg	0.896	0.696	0.740	322	
Mars + Earth Logistic Regression					
	precision	recall	f1-score	support	
safe	0.995	0.673	0.803	275	
hazard	0.338	0.979	0.503	47	
accuracy			0.717	322	
macro avg	0.666	0.826	0.653	322	
weighted avg	0.899	0.717	0.759	322	

Logistic Regression Classification Report

Mars Mini-CNN					
	precision	recall	f1-score	support	
safe	0.971	0.487	0.649	275	
hazard	0.234	0.915	0.372	47	
accuracy			0.550	322	
macro avg	0.602	0.701	0.511	322	
weighted avg	0.863	0.550	0.609	322	
Mars + Earth Mini-CNN					
	precision	recall	f1-score	support	
safe	0.919	0.902	0.910	275	
hazard	0.481	0.532	0.505	47	
accuracy			0.848	322	
macro avg	0.700	0.717	0.708	322	
weighted avg	0.855	0.848	0.851	322	

Mini-CNN Classification Report

Mars MobileNet					
	precision	recall	f1-score	support	
safe	0.948	0.935	0.941	275	
hazard	0.647	0.702	0.673	47	
accuracy			0.901	322	
macro avg	0.798	0.818	0.807	322	
weighted avg	0.904	0.901	0.902	322	
Mars + Earth MobileNet					
	precision	recall	f1-score	support	
safe	0.966	0.920	0.942	275	
hazard	0.633	0.809	0.710	47	
accuracy			0.904	322	
macro avg	0.799	0.864	0.826	322	
weighted avg	0.917	0.904	0.908	322	

MobileNet Classification Report